SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE
(AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)
Accredited by NAAC with 'A+' Grade.
Recognised as Scientific and Industrial Research Organisation
SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23	IV / IV - B.Tech. I - Semester									
COMPUTER SCIENCE AND DESIGN										
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)										
Course Code	**Course Name**	**Category**	**L**	**T**	**P**	**Cr**	**C.I.E.**	**S.E.E.**	**Total Marks**
B23CT4101	Fundamentals of Deep Learning	**PC**	3	0	0	3	30	70	100
B23HS4101	Human Resource & Project Management	**HS**	2	0	0	2	30	70	100
#PE-IV	Professional Elective-IV	**PE**	3	0	0	3	30	70	100
#PE-V	Professional Elective-V	**PE**	3	0	0	3	30	70	100
#OE-III	Open Elective-III	**OE**	3	0	0	3	30	70	100
#OE-IV	Open Elective-IV	**OE**	3	0	0	3	30	70	100
B23CD4109	Prompt Engineering	**SEC**	0	1	2	2	30	70	100
B23MC4101	Constitution of India	**MC**	2	0	0	0	30	-	30
B23CD4110	Evaluation of Industry Internship /Mini Project.	**PR**	-	-	-	2	-	50	50
		TOTAL	**19**	**01**	**02**	**21**	**240**	**540**	**780**

	Course Code	Course
# PE - IV	B23CS4102	Software Architecture & Design Patterns
	B23CD4101	Introduction to Block chain Technology
	B23CT4103	Data Science using Python
	B23IT4103	Computer Vision
	B23CD4102	Animation Principles and Design
	B23CD4103	12 week MOOC Swayam/ NPTEL course recommended by the BoS
# PE - V	B23CD4104	Introduction to Agile methodologies
	B23CD4105	Game Design & Development
	B23IT4108	Cyber Physical Systems
	B23CD4106	Digital Audio Design & Synthesis
	B23CD4107	Big Data Analysis
	B23CD4108	12 week MOOC Swayam/ NPTEL course recommended by the BoS
#OE-III	Student has to study one Open Elective offered by CE or ECE or EEE or ME or S&H from the list enclosed.	
#OE-IV	Student has to study one Open Elective offered by CE or ECE or EEE or ME or S&H from the list enclosed.	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CT4101	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

FUNDAMENTALS OF DEEP LEARNING

(Common to CSD & CSIT)

Course Objectives: The main objectives of the course is to provide students with:

1.	To understand Deep Learning concepts, techniques, and their real-world applications.
2.	To understand CNN algorithms and the way to evaluate performance of the CNN architectures.
3.	To apply RNN and LSTM models for prediction and classification in real-world problems.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Understand Deep learning concepts and Historical Trends	**K2**
2.	Explain the MP neuron and Perceptron learning for linearly separable problems	**K2**
3.	Illustrate the mathematical methods to train MLP, Forward and Backward Propagation	**K3**
4.	Implementing the basic of **C**onvolution operations and their role in DP models.	**K3**
5.	Analyze and implement advanced RNN architectures for sequence learning tasks.	**K4**

SYLLABUS

UNIT-I (10Hrs)	**Introduction:** Historical Trends in Deep Learning, The Many Names and Changing Fortunes of Neural Networks, Increasing Dataset Sizes, Increasing Model Sizes, Increasing Accuracy, Complexity and Real-World Impact.
UNIT-II (10 Hrs)	**Neuron & Neural Network:** The Brain And The Neuron: Hebb‘s Rule, McCulloch and PittsNeuronandItsLimitations,NeuralNetworks,ThePerceptron:TheLearningRate,BiasInput, ThePerceptron Learning Algorithm, Linear Separability, Linear Regression.
UNIT-III (10 Hrs)	**Multi-Layer Perceptron:** Going Forward: Biases, Back-Propagation and Error, The Multi-Layer Perceptron Algorithm, Initializing the Weights, Activation Functions, Sequential and Batch Training, LocalMinima, Picking up momentum, Mini batches and Stochastic Gradient Descent, Amount of Training Data, Number of Hidden Layers,Whento Stop Learning.
UNIT-IV (10 Hrs)	**Convolution Networks**: The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely StrongPrior, Variants of Basic Convolution Functions, Structured Outputs, Data Types, Efficient Convolution Algorithms, Random or Unsupervised Features, The Neuroscientific Basis for Convolutional Networks, Convolutional.
UNIT-V (10 Hrs)	**Sequence Modeling- Recurrent and Recursive Nets**: Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence Architectures, Deep Recurrent Networks, Recursive Neural Networks, The Challenge of

	Long-Term Dependencies, Echo State Networks, Leaky Units and Other Strategies for Multiple Time Scales, LSTM and Other Gated RNNs, Explicit Memory.	
Text Books:		
1.	Deep Learning, Ian Goodfellow, YoshuaBengio, and Aaron Courville, MIT Press. Reference Books:	
2.	Neural Networks and Deep Learning, Charu C. Aggarwal, Springer.	
Reference Books:		
1.	Fundamentals of Deep Learning, Nikhil Buduma, 1st Edition, O'Reilly	
e-Resources		
1.	https://www.oreilly.com/library/view/fundamentals-of-deep/9781491925607/	
2.	https://cs231n.github.io/optimization-2/	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS4101	HS	2	--	--	2	30	70	3 Hrs.

HUMAN RESOURCE & PROJECT MANAGEMENT

(Common to CSE, AIML, IT, AIDS, CSG, CSIT)

Course Objectives:

1.	Explain the nature, scope, functions of HRM, and processes of HR planning, recruitment, and selection.
2.	Explain HRD concepts, training methods, performance appraisal systems, and career development practices.
3.	Explain the basic concepts of project management, including project life cycle, resource management, and monitoring techniques.
4.	Explain project selection and appraisal methods using financial and strategic evaluation tools.
5.	Explain project implementation processes, organizational structures, and performance evaluation methods.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Explain HRM functions, HR planning, and recruitment–selection procedures.	K2
2.	Explain HRD practices, training techniques, and performance appraisal methods.	K2
3.	Explain project management concepts, project networks, and monitoring processes.	K2
4.	Explain project appraisal techniques such as SWOT analysis, payback period, and NPV.	K2
5.	Explain project implementation strategies, organizational forms, and performance evaluation processes.	K2

SYLLABUS

UNIT-I (10Hrs)	**HRM:** Nature, Scope, Concept of HRM - Functions of HRM - Role of HR manager – Emerging trends in HRM – E-HRM - HR audit models – ethical aspects of HRM. HR Planning – Job Design- Recruitment - Sources of recruitment -Selection- Selection Procedure.
UNIT-II (10 Hrs)	**HRD**: HR accounting – Models - Concept of Training and Development –- Methods of Training. Performance Appraisal: Importance – Methods of performance appraisal – Career Development and Counseling – group interaction.
UNIT-III (10 Hrs)	**Basics of Project Management:** Concept–resource management- Project environment – Types of Projects –project networks-DPR- Project life cycle – Project proposals – Monitoring project progress – Project appraisal and Project selection-80-20 rules.

UNIT-IV (10 Hrs)	**Project Selection and Appraisal:** Brainstorming and concept evolution, Project selection and evaluation, Selection criteria and models, Types of appraisals, SWOT analysis, Cash flow analysis, Payback period, and Net present value.
UNIT-V (10 Hrs)	**Project Implementation and Review:** Identify various project types and apply appropriate management strategies for each. Forms of project organization – project planning – project control – human aspects of project management – prerequisites for successful project implementation – project review.
Textbooks:	
1.	Sharon Pande and Swapnalekha Basak, Human Resource Management, Text and Cases, Vikas Publishing, 2e, 2016.
2.	K. Nagarajan, Project Management, New Age International Publishers, 8e, 2017
Reference Books:	
1.	Subba Rao P: ―Personnel and Human Resource Management-Text and Cases‖, Himalaya Publications, Mumbai, 2013.
2.	Prasanna Chandra, ―Projects, Planning, Analysis, Selection, Financing Implementation and Review‖, Tata McGraw Hill Company Pvt. Ltd., New Delhi 1998
3.	Vasanth Desai, Project Management, 4th edition, Himalaya Publications 2018.
4.	Lalithabalakrishnan and Gowri, Project Management, Himalaya publishing house, New Delhi 2022
5.	K Aswathappa: ―Human Resource and Personnel Management‖, Tata McGraw Hill, New Delhi, 2013
e-Resources	
1.	Robert L. Mathis, John H. Jackson, Manas Ranjan Tripathy, Human Resource Management, Cengage Learning 2016.
2.	Stewart R. Clegg, Torgeir Skyttermoen, Anne Live Vaagaasar, Project Management, Sage Publications, 1e, 2021.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4102	PE	3	--	--	3	30	70	3 Hrs.

SOFTWARE ARCHITECTURE & DESIGN PATTERNS

(Common to CSE, AIDS, CSBS, CSD)

Pre-requisites: OOP,SE,OOAD

Course Objectives: Students are expected

1.	Understand the basic concepts to identify state behaviour of real world objects
2.	Apply Object Oriented Analysis and Design concepts to solve complex problems
3.	Construct various UML models using the appropriate notation for specific problem context
4.	Design models to Show the importance of systems analysis and design in solving complex problems using case studies
5.	Study Pattern Oriented approach for real world problems

Course OutComes: At the end of the course students will be able to

S. No	**outcome**	KL
1.	**Explain** the concepts of design patterns and object-oriented development.	2
2.	**Use** object-oriented analysis techniques to gather requirements and identify conceptual classes and relationships in a system.	3
3.	**Demonstrate** structural design patterns such as Adapter, Bridge, Composite, Decorator, Facade, Flyweight, and Proxy.	3
4.	**Illustrate** the MVC architectural pattern in designing interactive systems.	3
5.	**Apply** distributed object system concepts including client–server architecture, Java RMI, and web services.	3

SYLLABUS

UNIT-I (10 Hrs)	Introduction: design pattern, Describing design patterns, the catalog of design pattern, organizing the catalog, how design patterns solve design problems, how to select a design pattern, how to use a design pattern. What is object oriented development-key concepts of object oriented design other related concepts, benefits and drawbacks of the paradigm
UNIT-II (10 Hrs)	Analysis a System: Overview of the analysis phase, stage 1 gathering the requirements functional requirements specification, defining conceptual classes and relationships, using the knowledge of the domain Design and Implementation, discussions and further reading
UNIT-III (10 Hrs)	Design Pattern Catalog: Structural patterns, Adapter, bridge, composite, decorator, facade, Flyweight, proxy.
UNIT-IV (10 Hrs)	Interactive systems and the MVC architecture: Introduction The MVC architectural pattern, analyzing a simple drawing program designing the system, designing of the subsystems, getting into implementation, implementing undo operation drawing incomplete items, adding a new feature pattern based solutions

UNIT-V (10 Hrs)	Designing with Distributed Objects: Client server system, java remote method invocation, implementing an object oriented system on the web, Web services (SOAP, Restful), Enterprise Service Bus

Text books:

1.	Object Oriented Analysis, Design and Implementation, Brahma Dathan, Sarnath Rammath , Universities Press, 2013
2.	Design Patterns, Erich Gamma, Richard Helan, Ralph Johman, John Vlissides, PEARSON Publication, 2013

Reference books:

1.	Frank Bachmann, Regine Meunier, Hans Rohnert "Pattern Oriented Software Architecture", Volume 1, 1996.
2.	William J Brown et al., "Anti Patterns: Refactoring Software, Architectures and Projects in Crisis", John Wiley, 1998

E-Resources:

1.	Object Oriented Analysis and Design By Prof. Partha Pratim Das, Prof. Samiran Chattopadhyay, Prof. Kausik Datta | IIT Kharagpur https://onlinecourses.nptel.ac.in/noc21_cs57/preview
2.	Object Oriented System Development Using UML, Java And Patterns, IIT Kharagpur Prof. Rajib Mall https://nptel.ac.in/courses/106105224

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CD4101	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

INTRODUCTION TO BLOCK CHAIN TECHNOLOGY
(For CSD)

Course Objectives: The main objectives of the course is to provide students with:

1.	To introduce the fundamentals and cryptographic foundations of block chain technology.
2.	To explain Bitcoin blockchain structure and the technologies underlying blockchain systems.
3.	To understand various consensus mechanisms used in blockchain networks.
4.	To study Ethereum architecture, smart contracts, and Solidity programming concepts.
5.	To explore Hyperledger Fabric, permissioned blockchains, and real-world blockchain applications

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Discuss the Cryptographic primitives used in Blockchain.	**K2**
2.	Discuss about various technologies borrowed in blockchain.	**K2**
3.	Illustrate various models for blockchain.	**K2**
4.	Discuss about Ethereum.	**K2**
5.	Discuss about Hyperledger Fabric.	**K2**

SYLLABUS

UNIT-I (10Hrs)	**INTRODUCTION TO BLOCKCHAIN:** Introduction, history of Bitcoin and origins of Blockchain, Fundamentals of Blockchain and key components (Chapter 1-book1), Permission and Permission-less platforms(Chapter 1-book2), Introduction to Cryptography, SHA256 and ECDSA, Hashing and Encryption, Symmetric/ Asymmetric keys, Private and Public Keys(Chapter 3-book2).
UNIT-II (10 Hrs)	**TECHNOLOGIES BORROWED IN BLOCKCHAIN:** Technologies Borrowed in Blockchain –hash pointers- - Digital cash etc.- Bitcoin blockchain - Wallet – Blocks Merkley Tree - hardness of mining - Transaction verifiability - Anonymity - forks - Double spending - Mathematical analysis of properties of Bitcoin - Bitcoin- the challenges and solutions. (Chapter 3-book2).
UNIT-III (10 Hrs)	**CONSENSUS MECHANISMS:** Consensus Algorithms: Proof of Work (PoW) as random oracle - Formal treatment of consistency- Liveness and Fairness - Proof of Stake (PoS) based Chains -Hybrid models (PoW + PoS), Byzantine Models of fault tolerance. ((Chapter 1- book2)).
UNIT-IV (10 Hrs)	**ETHEREUM:** Ethereum -Ethereum Virtual Machine (EVM) -Wallets for Ethereum - Solidity - Smart Contracts (Chapter 5-book1), - The Turing Completeness of Smart Contract Languages and verification challenges- Using smart contracts to enforce legal

	contracts- Comparing Bitcoin scripting vs. Ethereum Smart Contracts-Some attacks on smart contracts (Chapter 6 and Chapter 7-book2.
UNIT-V (10 Hrs)	**HYPERLEDGER FABRIC:** Hyperledger fabric- the plug and play platform and mechanisms in permissioned blockchain - Beyond Cryptocurrency – applications of blockchain in cyber security- integrity of information- E-Governance and other contract enforcement mechanisms - Limitations of blockchain as a technology and myths vs reality of blockchain technology (Chapter 16-book1), (Chapter 9 -book2).
Text Books:	
1.	Blockchain Technology Chandramouli Subramanian, Asha A George, Abhilash K A and Meena Karthikeyan, University Press, 2020.
2.	Mastering Blockchain - Distributed ledger technology, decentralization, and smart contracts explained, Imran Bashir,2nd ed. Edition,2018, pakct publication.
Reference Books:	
1.	S.Shukla, M.Dhawan, S.Sharma,S. Venkatesan “Blockchain Technology: Cryptocurrency and Applications” ,Oxford University Press 2019 .
2.	Cryptography and network security principles and practice, William Stallings, Pearson, 8th edition
3.	Blockchain Technology*Chandramouli Subramanian, Asha A George, Abhilash K A and MeenaKarthikeyan, Universities Press*
e-Resources	
1.	https://drive.google.com/file/d/1PtYaDmWYaqPVGjKDnMYGWO5eoI5wMPtJ/view
2.	https://www.tutorialspoint.com/blockchain/index.htm
3.	https://archive.nptel.ac.in/courses/106/104/106104220/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CT4103	PE	3	--	--	3	30	70	3 Hrs.

DATA SCIENCE USING PYTHON

(Common to CSD & CSIT)

Course Objectives: The main objectives of the course is to provide students with:

1.	Knowledge and expertise to become a data scientist.
2.	Essential concepts of statistics and machine learning that are vital for data science.
3.	Suitability and limitations of tools and techniques related to data science process

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Understand the basic concepts of Data Science.	**K2**
2.	Explain the Machine Learning Activities on Data Science	**K2**
3.	Describe databases and NoSQL technologies for handling big data.	**K2**
4.	Illustrate the tools and applications on data science.	**K3**
5.	Apply the Visualization and development tools for data.	**K3**

SYLLABUS

UNIT-I (10Hrs)	Introduction to Data science, benefits and uses, facets of data, data science process in brief, big data ecosystem and data science. **Data Science process:** Overview, defining goals and creating project charter, retrieving data, cleansing, integrating and transforming data, exploratory analysis, model building, presenting findings and building applications on top of them
UNIT-II (10 Hrs)	Applications of machine learning in Data science, role of ML in DS, Python tools like sklearn, modeling process for feature engineering, model selection, validation and prediction, types of ML, semi-supervised learning **Handling large data:** problems and general techniques for handling large data, programming tips for dealing large data, case studies on DS projects for predicting malicious URLs, for building recommender systems
UNIT-III (10 Hrs)	**NoSQL movement for handling Big data:** Distributing data storage and processing with Hadoop frame work, case study on risk assessment for loan sanctioning, ACID principle of relational databases, CAP theorem, base principle of NoSQL databases, types of NoSQL databases, case study on disease diagnosis and profiling.
UNIT-IV (10 Hrs)	**Tools and Applications of Data Science:** Introducing Neo4jfor dealing with graph databases, graph query language Cypher, Applications graph databases, python libraries like nltk and SQLite for handling Text mining and analytics, case study on classifying Reddit posts.

UNIT-V (10 Hrs)	**Data Visualization and Prototype Application Development:** Data Visualization options, Cross filter, the Java Script Map Reduce library, Creating an interactive dashboard with dc.js, Dashboard development tools. Applying the Data Science process for real-world problem-solving scenarios as a detailed case study.

Text Books:	
1.	Davy Cielen, Arno D.B. Meysman, and Mohamed Ali, "Introducing to Data Science using Python tools", Manning Publications Co, Dream tech press, 2016.
2.	Prateek Gupta, "Data Science with Jupyter" BPB publishers, 2019 for basics .
Reference Books:	
1.	Joel Grus "Data Science from Scratch", OReilly,2019
2.	Doing Data Science: Straight Talk from the Frontline,1st Edition, Cathy O' Neil and Rachel Schutt, O'Reilly, 2013
e-Resources	
1.	https://swayam.gov.in/nd1_noc19_cs60/preview
2.	https://towardsdatascience.com/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4103	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

COMPUTER VISION

(Common to IT&CSD)

Course Objectives: This course is designed to:

1.	To understand the Fundamental Concepts related to sources, shadows and shading
2.	To understand the Geometry of Multiple Views

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	**Explain** the fundamental principles of image formation and lighting	K2
2.	**Demonstrate** linear filtering techniques, frequency-domain concepts, edge detection methods, and texture representation models	K2
3.	**Apply** multi-view geometry concepts, stereoscopic reconstruction techniques, and clustering-based segmentation methods	K3
4.	**Apply** grouping and model fitting techniques, including Hough Transform and probabilistic methods	K3
5.	**Implement** geometric camera models, perform camera calibration using estimation techniques.	K3

SYLLABUS

UNIT- (10Hrs)	**CAMERAS:** Pinhole Cameras Radiometry – Measuring Light: Light in Space, Light Surfaces, Important Special Cases Sources, Shadows, And Shading: Qualitative Radiometry, Sources and Their Effects, Local Shading Models, Application: Photometric Stereo, Inter reflections: Global Shading Models Color: The Physics of Color, Human Color Perception, Representing Color, A Model for Image Color, Surface Color from Image Color.
UNIT-II (8 Hrs)	**Linear Filters:** Linear Filters and Convolution, Shift Invariant Linear Systems, Spatial Frequency and Fourier Transforms, Sampling and Aliasing, Filters as Templates, Edge Detection: Noise, Estimating Derivatives, Detecting Edges Texture: Representing Texture, Analysis (and Synthesis) Using Oriented Pyramids, Application: Synthesis by Sampling Local Models, Shape from Texture.
UNIT-III (10 Hrs)	**The Geometry of Multiple Views:** Two Views Stereopsis: Reconstruction, Human Stereopsis, Binocular Fusion, Using More Cameras Segmentation by Clustering: What Is Segmentation? Human Vision: Grouping and GetStalt, Applications: Shot Boundary Detection and Background Subtraction, Image Segmentation by Clustering Pixels, Segmentation by Graph-Theoretic Clustering.

UNIT-IV (10 Hrs)	**Segmentation by Fitting a Model:** The Hough Transform, Fitting Lines, Fitting Curves, Fitting as a Probabilistic Inference Problem, Robustness Segmentation and Fitting Using Probabilistic Methods: Missing Data Problems, Fitting, and Segmentation, The EM Algorithm in Practice, Tracking With Linear Dynamic Models: Tracking as an Abstract Inference Problem, Linear Dynamic Models, Kalman Filtering, Data Association, Applications and Examples
UNIT-V (8 Hrs)	**Geometric Camera Models:** Elements of Analytical Euclidean Geometry, Camera Parameters and the Perspective Projection, Affine Cameras and Affine Projection Equations Geometric Camera Calibration: Least-Squares Parameter Estimation, A Linear Approach to Camera Calibration, Taking Radial Distortion into Account, Analytical Photogrammetry, Case study: Mobile Robot Localization Model- Based Vision: Initial Assumptions, Obtaining Hypotheses by Pose Consistency, Obtaining Hypotheses by Pose Clustering, Obtaining Hypotheses Using Invariants, Verification, Case study: Registration In Medical Imaging Systems, Curved Surfaces and Alignment.
Textbooks:	
1.	David A. Forsyth and Jean Ponce: Computer Vision – A Modern Approach, PHI Learning (Indian Edition), 2009.
2.	Simon J.D Prince, Computer Vision: Models, Learning and Inference, 1st Edition, 2012.
Reference Books:	
1.	E. R. Davies: Computer and Machine Vision – Theory, Algorithms and Practicalities, Elsevier (Academic Press), 4th edition, 2013.
2.	R. C. Gonzalez and R. E. Woods “Digital Image Processing” Addison Wesley 2008. 3. Richard Szeliski “Computer Vision: Algorithms and Applications” Springer-Verlag London Limited 2011.
3.	Richard Szeliski “Computer Vision: Algorithms and Applications” Springer-Verlag London Limited 2011.
e-Resources	
1.	NPTEL LINK: https://onlinecourses.nptel.ac.in/noc22_ee48/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CD4102	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

ANIMATIONS PRINCIPAL AND DESIGN

(For CSD)

Course Objectives: The main objectives of the course is to provide students with:

1.	To understand the concept of 2D animation, cycles, and scenes.
2.	To understand basic concepts of animation, different types/ style and their workflow.
3.	To explain tools and techniques for 2D animation.
4.	To examine various processes of animation techniques that are developed with various equipment.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Explain the history, evolution and various types of animation techniques.	K2
2.	Develop scripts, storyboards, and character designs using script writing principles.	K3
3.	Apply animation tools for digital painting, vector drawing, and animation development.	K3
4.	Demonstrate animation principles, storytelling, editing, in animation production.	K3
5.	Apply advanced 2D animation, post-production, compositing, and effects techniques.	K3

SYLLABUS

UNIT-I (10Hrs)	**Drawing Techniques:** What is Animation, History of Animation – Starting from Early approaches to motion in art, Animation before film, Traditional Animation – The silent era, Walt Disney & Warner Bros., Snow White & the seven dwarfs, The Television era, Stop-motion, CGI Animation - till date. Different Types of Animation, Basic Drawing Concepts of Visualization Illustration and Sketching Basic Shapes and Sketching Techniques
UNIT-II (10 Hrs)	**Script Writing and Character Enhancement:** Basic Script and Story Board, Concepts of 2D, Cel Animation, Character Design and Development, Traditional animation, key frame animation, key poses and time stretch, character design development, facial expressions and walk cycles.
UNIT-III (10 Hrs)	**Animation Tools Processing:** Scripting & Storyboarding with Toon Boom Pro, Animation Process Development, Usage of tools for Digital Painting and vector drawings, How to develop a character and background creation, Usage of timeline and its purpose.
UNIT-IV	**Principles, Storytelling &Editing :** Animation Concept Acting and Direction for

(10 Hrs)	Animation Timing for Animation Storytelling Techniques Script Writing Concept Design and Development Storyboarding and Animatics, Audio and Video Streaming and Editing, Previsualization.
UNIT-V (10 Hrs)	**Advanced Techniques of Production, Digital Animation:** Advanced 2D Production and Post Production, Digital Animation Scene Planning, Digital Animation Ink and Paint, and Digital Animation Composting and Effects.
Textbooks:	
1.	Edoux, Trish, Ranney, “Complete Anime Guide: Japanese Animation Film Directory and Resource Guide”, ‘Tiger Mountain Press, 1997.
2.	Kevin Hedgpeth (Author), Stephen Missal, “Exploring Drawing for Animation ,Design Concepts,1st Edition, march 15, 2004.
3.	The Illusion of Life: Disney Animation - Frank Thomas and Ollie Johnston
Reference Books:	
1.	Pakhira Malay K, "Computer Graphics, Multimedia and Animation”, Second Edition, 2010.
2.	Preston Blair, "Cartooning: Animation 1 with Preston Blair: Learn to animate cartoons step by step (How to Draw & Paint)", Walter Foster Publishing, 2003
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/cec20_cs08/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CD4104	PE	3	--	--	3	30	70	3 Hrs.

INTRODUCTION TO AGILE METHODOLOGIES

(For CSD)

Course Objectives: The main objectives of the course is to provide students with:

1.	Understand software engineering principles and SDLC models.
2.	Apply Agile methodologies in real-world development. Learn requirement analysis and design techniques
3.	Understand testing, maintenance, and quality assurance.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Explain the evolution of software engineering, software crisis, waterfall model.	K2
2.	Discuss the requirements Elicitation Techniques, UML Diagrams, Design Process	K2
3.	Demonstrate Build and Fix Errors, Debugging Process, Debugging Techniques.	K3
4.	Apply the Agile Methodologies, Scrum and Extreme Programming (XP)	K3
5.	Evaluate project quality and agile metrics.	K4

SYLLABUS

UNIT-I (10Hrs)	**SOFTWARE ENGINEERING FUNDAMENTALS** **Evolution**: Evolution of Software Engineering, Software Crisis, **SDLC models:** Waterfall, Spiral, Agile, Project planning, Software Development Life Cycle (SDLC), Phases of SDLC, Overview of Process Models, Waterfall Model, Spiral Model, Agile Development Model, Comparison of SDLC Models, Project Planning, Scope Management, Effort Estimation Techniques (LOC, Function Point, COCOMO), Project Scheduling (Gantt Charts, PERT), Risk Management
UNIT-II (10 Hrs)	**REQUIREMENTS ENGINEERING & DESIGN:** **Requirement Engineering** – Introduction and Process, Feasibility Study, Requirements Elicitation Techniques (Interviews, Questionnaires, Observation, Prototyping), Requirements Analysis and Negotiation, Requirements Specification , Requirements Validation, Functional and Non-Functional Requirements **UML:** (Unified Modeling Language) – Introduction, Diagrams Overview, Use Case Diagram, Class Diagram, Sequence Diagram, Activity Diagram, State Chart Diagram, Collaboration Diagram **Software Design:** Concepts and Principles ,Design Process and Design Models ,Modularity and Abstraction ,Cohesion and Coupling ,Architectural Design Concepts ,StructuredDesign ,Object-Oriented Design Concepts

UNIT-III (10 Hrs)	**TESTING, DEBUGGING & MAINTENANCE:** **Software Testing** – Introduction and Objectives ,Verification and Validation **Levels of Testing** (Unit Testing, Integration Testing, System Testing, Acceptance Testing),**Types of Testing** (Black Box Testing, White Box Testing, Grey Box Testing) ,Functional Testing and Non-Functional Testing ,RegressionTesting ,Smoke Testing and Sanity Testing ,Test Case Design Techniques **Code Coverage Techniques: Debugging** – Introduction **,**DebuggingProcess Types of Errors (Syntax Errors, Logical Errors, Runtime Errors) ,Debugging Techniques (Brute Force, Backtracking, Cause Elimination) ,Tools and Practices for Debugging
UNIT-IV (10 Hrs)	**AGILE METHODOLOGIES:** **Agile Methodology** – Introduction ,Limitations of Traditional Models ,Agile Manifesto – Values and Principles ,Agile Process Models Overview **Scrum Framework** – Introduction **Scrum Roles** (Product Owner, Scrum Master, Development Team) ,**Scrum Artifacts** (Product Backlog, Sprint Backlog, Increment) **Scrum Events** (Sprint, Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective) Scrum Workflow **Extreme Programming** (XP) – Introduction , XP Values and Principles XP Practices (Pair Programming, Continuous Integration, Test-Driven Development, Refactoring, Small Releases) .**Lean Software Development** – Introduction ,Lean Principles (Eliminate Waste, Amplify Learning, Decide as Late as Possible, Deliver as Fast as Possible, Empower Team, Build Integrity, Optimize Whole)Comparison of Agile Methods (Scrum, XP, Lean).
UNIT-V (10.Hrs)	**AGILE PRACTICES:** User Stories – Introduction ,User Story Format and Structure ,Writing Effective User Stories ,Acceptance Criteria ,Story Prioritization Techniques ,Sprint Planning – Introduction Sprint Goals ,Backlog Refinement Task Breakdown and Estimation (Planning Poker, Story Points), Sprint Execution and Tracking Test-Driven Development (TDD) – Introduction ,TDD Cycle (Red–Green–Refactor) Writing Unit Tests Benefits and Limitations of TDD Continuous Integration and Continuous Deployment (CI/CD) – Introduction CI/CD Pipeline Build Automation ,Version Control Integration ,Automated Testing and Deployment Agile Metrics – Introduction Velocity ,Burndown and Burnup Charts

Text Books:

1.	Fundamentals of Software Engineering, Rajib Mall, 5th Edition PHI
2.	Agile Software Development, Principles, Patterns, and Practices (Alan Apt Series), Robert Martin

Reference Books:

1.	Ian Sommerville – Software Engineering, 10th Edition, Pearson Publications
2.	Alistair Cockburn – Agile Software Development.

e-Resources

1.	[NPTEL : Software Engineering (Computer Science and Engineering)]

2.	nptel.ac.in/courses/110104073

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CD4105	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

GAME DESIGN AND DEVELOPMENT

(For CSD)

Course Objectives: The main objectives of the course is to provide students with:

1.	Introduce fundamental concepts and practices in video game design and development.
2.	Explore game genres, storytelling, character and level design, and game development challenges.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Explain game development components and structure, and video games.	**K2**
2.	Illustrate gaming platforms and their impact on game design and user experience.	**K2**
3.	Implement game concepts into complete designs with creative worlds and expressive mechanics.	**K3**
4.	Create engaging characters and avatars with visual, narrative, and audio elements.	**K4**
5.	Analyze game storytelling techniques, balancing linear and nonlinear narratives using tools like dialogue trees.	**K4**

SYLLABUS

UNIT-I (10Hrs)	**Games and Video Games:** Game, Conventional Games Versus Video Games, Games for Entertainment, Serious Games, Designing and Developing Games: An Approach to the Task, Key Components of Video Games, The Structure of a Video Game, Stages of the Design Process, Game Design Team Roles, Game Design Documents, The Anatomy of a Game Designer.
UNIT-II (10 Hrs)	**The Major Genres:** Genre. The Classic Game Genres. Understanding Your Player: Vanden Berghe‘s Five Domains of Play, Demographic Categories, Gamer Dedication, The Dangers of Binary Thinking Understanding Your Machine: Home Game Consoles, Personal Computers, Portable Devices Other Devices
UNIT-III (10 Hrs)	**Game Concepts:** Getting an Idea, From Idea to Game Concept Game Worlds: What Is a Game World? The Purposes of a Game World, The Dimensions of a Game Creative and Expressive Play: Self-Defining Play, Creative Play, Other Forms of Expression, Game Modifications. Character **Development:** The Goals of Character Design, The Relationship Between Player and Avatar, Visual Appearances, Character Depth, Audio Design
UNIT-IV	**Storytelling:** Why Put Stories in Games? Key Concepts, The Storytelling Engine, Linear

(10 Hrs)	Stories, Nonlinear Stories, Granularity, Mechanisms for Advancing the Plot, Emotional Limits of Interactive Stories, Scripted Conversations and Dialogue Trees, When to Write the Story, Other Considerations
UNIT-V (10 Hrs)	**General Principles of Level Design:** What Is Level Design? Key Design Principles, Layouts, Expanding on the Principles of Level Design, The Level Design Process, Pitfalls of Level Design Design Issues for Online Gaming: What Are Online Games? Advantages of Online Games, Disadvantages of Online Games, Design Issues, Technical Security, Persistent Worlds, Social Problems
Text Books:	
1.	Fundamentals of Game Design, Third Edition, by Ernest Adams, Released December 2013,Publisher(s): New Riders, ISBN: 9780133435726
Reference Books:	
1.	Designing Games, A Guide to Engineering Experiences by Tynan Sylvester.
2.	Game Design Essentials by Briar Lee Mitchell.
e-Resources	
1.	NPTEL :NOC:Introduction to Game Theory and Mechanism Design (Computer Science and Engineering)
2.	Game Dev/Design Resources and Ideas | Video Game Design/Developers

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4108	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

CYBER PHYSICAL SYSTEMS

(Common to IT, CSE, CSD)

Course Objectives: This course is designed to:

1.	To introduce principles of cyber-physical systems, including embedded sensing, actuation, and control.
2.	To teach formal methods for modeling dynamic, real-time systems, such as state- space models and hybrid automata
3.	To enable students to apply AI techniques, control theory, and data analytics in practical engineering problems.

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	**Understand** and apply symbolic synthesis techniques to model, analyze, and design controllers for Cyber-Physical Systems	K3
2.	**Analyze** security requirements, attack models, and countermeasures in Cyber-Physical Systems,	K3
3.	**Understand** and address synchronization challenges in distributed Cyber-Physical Systems	K2
4.	**Apply** real-time scheduling techniques for Cyber Physical Systems	K3
5.	**Understand** and apply model integration techniques in Cyber-Physical Systems,	K3

SYLLABUS

UNIT-I	**Symbolic Synthesis for Cyber-Physical Systems:** Introduction and Motivation, Basic Techniques - Preliminaries, Problem Definition, Solving the Synthesis Problem, Construction of Symbolic Models, **Advanced Techniques:** Construction of Symbolic Models, Continuous Time Controllers, Software Tools
UNIT-II	**Security of Cyber-Physical Systems:** Introduction and Motivation, Basic Techniques - Cyber Security Requirements, Attack Model, Countermeasures, **Advanced Techniques:** System Theoretic Approaches
UNIT-III	**Synchronization in Distributed Cyber-Physical Systems:** Challenges in Cyber-Physical Systems, A Complexity-Reducing Technique for Synchronization, Formal Software Engineering, Distributed Consensus Algorithms, Synchronous Lockstep Executions, TimeTriggered Architecture, Related Technology, Advanced Techniques

UNIT-IV	**Real-Time Scheduling for Cyber-Physical Systems:** Introduction and Motivation, Basic Techniques – Scheduling with Fixed Timing Parameters, Memory Effects, Multiprocessor/Multicore Scheduling, Accommodating Variability and Uncertainty
UNIT-V	**Model Integration in Cyber-Physical Systems:** Introduction and Motivation, Causality, Semantic Domains for Time, Interaction Models for Computational Processes, Semantics of CPS DSMLs, Advanced Techniques, ForSpec, The Syntax of CyPhyML, Formalization of Semantics, Formalization of Language Integration.
Textbooks:	
1.	Raj Rajkumar, Dionisio De Niz, and Mark Klein, Cyber-Physical Systems, AddisonWesley Professional, 2016
2.	Rajeev Alur, Principles of Cyber-Physical Systems, MIT Press
Reference Books:	
1.	E.A.Lee, Sanjit Seshia, Introduction to Embedded Systems: A Cyber-Physical Systems Approach, MIT Press
2.	Andre Platzer, Logical Foundations of Cyber-Physical Systems, (2e), Springer Publishing, 2018

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CD4106	PE	3	--	--	3	30	70	3 Hrs.

DIGITAL AUDIO DESIGN & SYNTHESIS
(For CSD)

Course Objectives: The main objectives of the course is to provide students with:

1.	To learn about discrete time sampling, quantization, and signal processing.
2.	To understand and utilize general digital audio processing theory.
3.	Describe digital-to-analog and analog-to-digital conversion using PCM and explain digital audio creation.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Understand the basics of sound, digital audio and audio recording techniques.	K2
2.	Explain digital audio reproduction processes and digital-to-analog conversion techniques.	K2
3.	Explain error detection and correction techniques in digital audio systems.	K2
4.	Apply digital sound synthesis methods such as additive, and subtractive synthesis.	K3
5.	Apply advanced sound synthesis techniques such as modulation and stochastic synthesis.	K3

SYLLABUS

UNIT-I (10Hrs)	**Sound and Numbers:** Physics of Sound, Digital Basics, Binary Codes, Boolean Algebra, Analog versus Digital. **Fundamentals of Digital Audio:** Discrete Time Sampling, The Sampling Theorem, Aliasing, Quantization, Dither. Digital Audio Recording: Pulse-Code Modulation, Dither Generator, Input Lowpass Filter, Sample and-Hold Circuit, Analog-to-Digital Converter, Record Processing, Channel Codes.
UNIT-II (10 Hrs)	**Digital Audio Reproduction:** Reproduction Processing, Digital-to-Analog Converter, Output Sample and-Hold Circuit, Output Low pass Filter, Impulse Response, Digital Filters, Noise Shaping, Output Processing, Alternate Coding Architectures, Time base Correction.
UNIT-III (10 Hrs)	**Error Correction:** Sources of Errors, Quantifying Errors, Objectives of Error Correction, Error Detection, Error-Correction Codes, Reed-Solomon Codes, CIRC, Product Codes, Error Concealment.
UNIT-IV (10 Hrs)	**Digital Sound Synthesis:** Introduction to Digital Sound Synthesis, Sampling and Additive Synthesis, Multiple Wavetable, Wave Terrain, Granular, and Subtractive Synthesis.

UNIT-V (10 Hrs)	**Digital Sound Synthesis:** Modulation Synthesis, Physical Modelling and Format Synthesis, Waveform Segment, Graphic, and Stochastic Synthesis.
Text Books:	
1.	Ken C. Pohlmann, Principles of Digital Audio, Sixth Edition, O'Reilly.
2.	The Computer Music Tutorial by Curtis Roads.
Reference Books:	
1.	On Sonic Art by Trever Wishart Andy Farnell, Designing Sound, 2010, ISBN: 9780262014410
2.	Francis Rumsey & Tim McCormick "Sound and Recording ", Sixth Edition, 2009, Focal Press, Elsevier Ltd.
e-Resources	
1.	https://www.my-mooc.com/en/mooc/digitalsounddesign/
2.	https://www.tensorflow.org/io/tutorials/audio
3.	https://www.dspguide.com/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CD4107	PE	3	--	--	3	30	70	3 Hrs.

BIG DATA ANALYTICS

(Common to CSD & CSIT)

Course Objectives: The main objectives of the course is to provide students with:

1.	Provide an overview of an exciting growing field of Big Data analytics.
2.	Understand how to solve complex problems using Map Reduce
3.	Introduce the tools required to analyze Big Data like Hadoop Map Reduce, Pig & Hive etc.,

Course Outcomes:

S.No	Outcome	Knowledge Level
1.	Understand technologies and the need for file systems in Big Data analysis.	K2
2.	Explain the features of HDFS, Map Reduce to handle the Big Data.	K2
3.	Implement and analyze Map-Reduce programming model for better optimization.	K3
4.	Apply stream processing techniques for real-time data analysis in the Spark environment.	K3
5.	Interpret the need of Modern tools, viz., Pig and Hive, Hbase etc., on Big Data.	K3

SYLLABUS

UNIT-I (10Hrs)	**Data structures in Java:** Linked List, Stacks, Queues, Sets, Maps; Generics: Generic classes and Type parameters, Implementing Generic Types, Generic Methods, Wrapper Classes, Concept of Serialization
UNIT-II (10 Hrs)	**Working with Big Data:** Google File System, Hadoop Distributed File System (HDFS) Building blocks of Hadoop (Namenode, Datanode, Secondary Namenode, Job Tracker, Task Tracker), Introducing and Configuring Hadoop cluster (Local, Pseudo-distributed mode, Fully Distributed mode), Configuring XML files.
UNIT-III (10 Hrs)	**Writing Map Reduce Programs:** A Weather Dataset, Understanding Hadoop API for Map Reduce Framework (Old and New), Basic programs of Hadoop Map Reduce: Driver code, Mapper code, Reducer code, Record Reader, Combiner, Practitioner
UNIT-IV (10 Hrs)	**Stream Memory and Spark:** Introduction to Streams Concepts– Stream Data Model and Architecture , Stream computing, Sampling Data in a Stream , Filtering Streams ,Counting Distinct Elements in a Stream , Introduction to Spark Concept , Spark Architecture and components , Spark installation , Spark RDD(Resilient Distributed Dataset) – Spark RDD operations

UNIT-V (10 Hrs)	**Pig:** Hadoop Programming Made EasierAdmiring the Pig Architecture, Going with the Pig Latin Application Flow, Working through the ABCs of Pig Latin, Evaluating Local and Distributed Modes of Running Pig Scripts, Checking out the Pig Script Interfaces, Scripting with Pig Latin. **Applying Structure to Hadoop Data with Hive:** Saying Hello to Hive, Seeing How the Hive is Put Together, Getting Started with Apache Hive, Examining the Hive Clients, Working with Hive Data Types, Creating and Managing Databases and Tables, Seeing How the Hive Data Manipulation Language Works, Querying and Analysing data

Text Books:	
1.	Wiley & Big Java 4th Edition, Cay Horstmann, Wiley John Sons, INC
2.	Hadoop: The Definitive Guide by Tom White, 3rd.Edition, O'reilly
Reference Books:	
1.	Hadoop in Action by Chuck Lam, MANNING Publ.
2.	Hadoop for Dummies by Dirk deRoos, Paul C.Zikopoulos, Roman B.Melnyk,Bruce Brown, Rafael Coss
3.	Hadoop in Practice by Alex Holmes, MANNING Publ.
4.	Big Data Analytics by Dr. A.Krishna Mohan and Dr.E.Laxmi Lydia
5.	Hadoop Map Reduce Cookbook, SrinathPerera, ThilinaGunarathne
e-Resources	
1.	Hadoop: https://hadoop.apache.org/
2.	Hive:https://cwiki.apache.org/confluence/display/Hive/Home/
3.	Piglatin: https://pig.apache.org/docs/r0.7.0/tutorial.html

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CD4109	**SEC**	**--**	**1**	**2**	**2**	**30**	**70**	**3 Hrs.**

PROMPT ENGINEERING

(For CSD)

Course Objectives: The main objectives of the course is to provide students with:

1.	Apply iterative prompting for clarity and context.
2.	Create varied prompts to steer model outputs.
3.	Construct chain-of-thought and structured prompts.
4.	Develop retrieval-augmented pipelines to ground outputs.
5.	Evaluate LLM agents and multimodal apps for ethics and robustness.

Course Outcomes:

S.No	Outcome	Knowledge Level
1.	Apply prompting techniques for classification, question answering, and code generation.	**K3**
2.	Design prompts for text generation, debugging, SQL queries, chatbots.	**K3**
3.	Construct advanced prompts for structured outputs and real-world problem solving.	**K3**

SYLLABUS

UNIT-I (10Hrs)	**Foundations of Prompt Engineering:** Definition of prompt engineering, Distinction between prompt engineering and model fine-tuning, Motivation and benefits of prompt engineering, Core principles of effective prompt design**,** Anatomy of a prompt, Setting up the Python environment for LLM interaction, Iterative prompting lifecycle, Common prompt pitfalls and remediation **Lab Experiments:** 1. Environment & Connectivity: Install required packages (e.g., transformers, openai); securely configure the API key; run a simple “Hello, world” prompt to verify model access. 2. Baseline vs. Enhanced Prompts: Execute a naïve prompt (“Write a one-paragraph bio of Ada Lovelace.”) and an enhanced prompt that adds role framing, specificity, and explicit format instructions; compare both outputs for relevance, completeness, and style. 3. Iterative Refinement on a Simple Task: Summarize the plot of the Shakespearean play Romeo and Juliet in two sentences through three rounds of prompt tweaking: a. Minimal instruction. b. Addition of length and style constraints c. Specification of key content elements (setting and theme) Document how each iteration changes and improves the result. 4. Diagnosing Prompt Failures & Edge Cases: Craft a vague or contradictory prompt;

	analyze the failure mode (ambiguity, missing context, or format errors); refine the prompt by adding examples or clarifying instructions.
UNIT-II (10 Hrs)	**Advanced Prompt Patterns & Techniques:** Enhanced prompt anatomy: contextual detail and explicit output specifications, Few-shot in-context prompting, Prompt structuring and template design, Role-based prompting to establish personas or system behavior, Negative prompting to filter or suppress undesired content, Constraint specification and instruction enforcement (e.g., length, format), Iterative prompt refinement and optimization. **Lab Experiments:** 1. Few-Shot vs. Zero-Shot Comparison: Design and execute a zero-shot prompt and a few-shot prompt (with 2–3 exemplar input-output pairs) for a chosen text task (e.g., sentiment 2. classification or translation); compare outputs for accuracy, consistency, and adherence to examples. 3. Role-Based & Negative Prompting: Craft a role-based prompt to establish a specific persona (e.g., "You are a financial advisor…"); then create a negative prompt to suppress undesired content (e.g., "Do not mention any brand names"); evaluate how each influences the model's response. 4. Constraint Specification & Iterative Refinement: Select an open-ended task (e.g., summarizing a technical article); issue a basic prompt; identify failures in length or format; refine the prompt by adding explicit constraints (word count, bullet format, etc.); document improvements over two refinement cycles.
UNIT-III (10 Hrs)	**Unit III: Structured Output & Reasoning Techniques:** Importance of structured outputs for real-world applications, Prompting for specific formats (lists, tables, Markdown), Generating valid JSON and YAML via explicit instructions, Eliciting chain-of-thought reasoning in zero-shot prompts, Decomposing complex tasks into manageable sub-tasks **Lab Experiments:** 1. Structured Format Prompting: Instruct the model to output information as bullet lists and Markdown tables (e.g., "List three benefits of daily exercise in a Markdown table with columns 'Benefit' and 'Description.'"); verify the output matches the requested structure. 2. JSON/YAML Generation: Provide a brief dataset description (e.g., three books with title, author, publication year) and prompt the model to produce valid JSON or YAML; use a parser to validate syntax and refine the prompt if errors occur. 3. Chain-of-Thought & Task Decomposition: Present a multi-step problem (e.g., a logic puzzle) and apply zero-shot CoT prompting (e.g., "Let's think step by step. Explain your reasoning before the final answer."); separately, decompose the problem into sequential sub-questions, collect partial answers, combine them, and compare accuracy against a direct-answer baseline.

UNIT-IV (10 Hrs)	**Unit IV: Retrieval-Augmented Generation &LangChain Workflows:** Limitations of LLM internal knowledge, Need for external data sources, Introduction to Retrieval-Augmented Generation (RAG), Overview of RAG architecture (indexing vs. retrieval + generation), Getting started with LangChain for LLM applications, Basics of LangChain Expression Language (LCEL), Simplified indexing pipeline: document loading & text splitting, Fundamentals of embeddings and vector stores, Building a basic retrieval-generation pipeline with an LCEL chain **Lab Experiments:** 1. Building a Simple LCEL Chain: Create a minimal LCEL script that accepts a fixed instruction (e.g., "Summarize this text: …"), passes it to an LLM, and prints the result; verify end-to-end execution. 2. Basic Data Indexing for RAG: Load a small collection of documents; split into uniform chunks (e.g., 200 tokens); generate embeddings for each chunk; store them in an in-memory vector store; inspect for consistency. 3. Constructing & Running a Basic RAG Chain: Build a pipeline that: a. Receives a user query b. Retrieves the top-k relevant chunks c. Constructs a combined prompt with context + query d. Send it to the LLM e. Returns the answer Test with sample queries and compare factual accuracy against a prompt without retrieval.
UNIT-V (10 Hrs)	**Agents, Multimodal AI & Ethical Evaluation:** Introduction to LLM agents and their basic architecture, Overview of multimodal AI models (VLMs), Prompting for text-to-image generation and image understanding, Importance of prompt evaluation beyond subjective judgment, Manual evaluation techniques (heuristic checks for accuracy, relevance, format), Introduction to "LLM-as-Judge" for automated evaluation, Security considerations (prompt injection, sensitive-information risks), Prompt-based mitigation strategies for safety and robustness, Ethical concerns (bias, misinformation, data privacy), Brief exploration of UI frameworks (Streamlit/Gradio) for deploying prompt-driven apps, Adapting to the evolving nature of prompt engineering through continuous learning **Lab Experiments:** 1. Building a Simple LLM Agent: Register a tool (e.g., a calculator function) and craft prompts that instruct the agent to invoke it when required; implement using LangChain or a function-calling API; test on queries requiring tool execution. 2. Multimodal Prompting Exploration: Generate images from detailed text prompts; feed one generated image into an image-understanding model or API with an appropriate prompt; compare the returned caption to the original prompt to evaluate alignment. 3. Prompt Evaluation & Ethics Workshop: a. Select two existing prompts and generate multiple outputs; apply manual heuristic checks for accuracy, relevance, and format compliance. b. Use an "LLM-as-Judge" prompt (e.g., "Rate these outputs on a scale of 1–5 for clarity and correctness.") to automate evaluation.

	c. Design a prompt- injection test (e.g., "Ignore previous instructions…"), observe the response, then refine system prompts to mitigate the vulnerability.
Text Books:	
1.	Prompt Engineering for Generative AI, James Phoenix, Mike Taylor, 1st Edition, O'Reilly Media
2.	Generative AI on AWS, Chris Fregly, Antje Barth, 1st Editio, O'Reilly Media
Reference Books:	
1.	Brown, T. B., et al. *"Language Models are Few-Shot Learners."* Advances in Neural Information Processing Systems (NeurIPS), 2020.
2.	Russell, S., & Norvig, P. *"Artificial Intelligence: A Modern Approach."* 4th Edition, Pearson, 2020.
e-Resources	
1.	Preview: Secure Computation: Part I | SWAYAM
2.	LangChain overview - Docs by LangChain
3.	Prompt engineering | OpenAI API

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4101	**MC**	**2**	**--**	**--**	**--**	**30**		**--**

CONSTITUTION OF INDIA

(Common to all)

Course Objectives: The main objectives of the course is to provide students with:

1.	To understanding of the fundamental principles, and key features of the Indian Constitution.
2.	To familiarize students with the structure and functioning of Union, State, Governments.
3.	To Develop awareness of constitutional bodies and welfare commissions.

Course Outcomes: At the end of the course, students will be able to

S.No.	Outcome	Knowledge Level
1.	Understand the concepts of the Indian Constitution.	K2
2.	Explain the structure and functions of the Union Government.	K2
3.	Explain the structure, functions of State Government and administrative roles.	K2
4.	Examine local administration and the roles of district, municipal, and Panchayati	K3
5.	Demonstrate the roles of the Election Commission and welfare commissions in governance and social justice.	K3

SYLLABUS

UNIT-I (6 Hrs)	Introduction to Indian Constitution: Constitution meaning of the term, Indian Constitution - Sources and constitutional history, Features - Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.
UNIT-II (6Hrs)	Union Government and its Administration Structure of the Indian Union: Federalism, Centre- State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha, The Supreme Court and High Court: Powers and Functions.
UNIT-III (6Hrs)	State Government and its Administration Governor - Role and Position - CM and Council of ministers, State Secretariat: Organisation, Structure and Functions.
UNIT-IV (6Hrs)	Local Administration - District's Administration Head - Role and Importance, Municipalities — Mayor and role of Elected Representative - CEO of Municipal Corporation Pachayati Raj: Functions PRI: Zilla Panchayat, Elected officials and their roles, CEO Zilla Panchayat: Block level Organizational Hierarchy - (Different departments), Village level - Role of Elected and Appointed officials - Importance of grass

	root democracy.
UNIT-V (6Hrs)	Election Commission: Election Commission- Role of Chief Election Commissioner and Election Commissionerate State Election Commission: Functions of Commissions for the welfare of SC/ST/OBC and women.

Textbooks:	
1.	Durga Das Basu, Introduction to the Constitution of India, Prentice — Hall of India Pvt. Ltd..New Delhi.
2.	Subash Kashyap, Indian Constitution, National Book Trust.
Reference Books:	
1.	J.A. Siwach, Dynamics of Indian Government & Politics, Sterling Publishers.
2.	D.C. Gupta, Indian Government and Politics, S Chand.
3.	H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes, Universal Law Publication.
4.	M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice Hall of India Pvt. Ltd.. New Delhi.
5.	Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012.
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg151
2.	https://onlinecourses.nptel.ac.in/noc24_lw05/preview
3.	https://nptel.ac.in/courses/129106750

Regulation: R23	IV / IV - B.Tech. II - Semester								
COMPUTER SCIENCE AND DESIGN									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2023-24 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**L**	**T**	**P**	**Cr**	**C.I.E.**	**S.E.E.**	**Total Marks**
B23CD4201	Project and Internship	**PR**	**-**	**-**	**24**	**12**	**60**	**140**	**200**
	TOTAL		**-**	**-**	**24**	**12**	**60**	**140**	**200**

Course Code	Category	L	T	P	C	I.M.	E.M.	Exam
B23CD4201	**PR**	**--**	**--**	**24**	**12**	**60**	**140**	**3H**

PROJECT WORK

Pre-requisites:: SE and Technical Skills in different Technologies, skills, planning, documentation

Course Objectives: The main objectives of the course are to

1.	To provide an opportunity to work in group on a topic / problem / experimentation
2.	To encourage creative thinking process
3.	To provide an opportunity to analyze and discuss the results to draw conclusions
	To acquire and apply fundamental principles of planning and carrying out the work plan of the project through observations, discussions and decision-making process

Course Outcomes: At the end of the course, students will be able to

S.No.	Outcome	Knowledge Level
1.	Identify a current problem through literature/field/case studies	K3
2.	Identify the objectives and methodology for solving the problem	K3
3.	Design and Develop technology/process for solving the problem	K4
4.	Evaluate the technology/process	K5

SYLLABUS

*The object of Project Work is to enable the student to take up investigative study in the broad field of Computer Science and Design, either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a Supervisor.

This is expected to provide a good initiation for the student(s) in R&D work.

The assignment to normally include:

a) Survey and study of published literature on the assigned topic.

b) Working out a preliminary approach to the problem relating to the assigned topic.

c) Conducting preliminary Analysis/Modelling/Simulation/ Experiment/ Design / Feasibility.

d) Preparing a written report on the study conducted for presentation to the department.

e) Final Seminar, as oral Presentation before a departmental committee